

Title: Program to implement Circular Queue.

```
#include <iostream>

using namespace std;

int cqueue[5];

int front = -1, rear = -1, n = 5;

void insertCQ(int val) {
    if ((front == 0 && rear == n - 1) || (front == rear + 1)) {
        cout << "Queue Overflow\n";
        return;
    }

    if (front == -1) {
        front = 0;
        rear = 0;
    }
    else {
        if (rear == n - 1)
            rear = 0;
        else
            rear = rear + 1;
    }

    cqueue[rear] = val;
}

void deleteCQ() {
    if (front == -1) {
```

```
    cout << "Queue Underflow\n";  
    return;  
}
```

```
cout << "Element deleted from queue is : " << cqueue[front] << endl;
```

```
if (front == rear) {  
    front = -1;  
    rear = -1;  
}  
else {  
    if (front == n - 1)  
        front = 0;  
    else  
        front = front + 1;  
}  
}
```

```
void displayCQ() {  
    int f = front, r = rear;  
  
    if (front == -1) {  
        cout << "Queue is empty" << endl;  
        return;  
    }
```

```
    cout << "Queue elements are :\n";
```

```
    if (f <= r) {  
        while (f <= r) {  
            cout << cqueue[f] << " ";
```

```

        f++;
    }
}
else {
    while (f <= n - 1) {
        cout << cqueue[f] << " ";
        f++;
    }

    f = 0;

    while (f <= r) {
        cout << cqueue[f] << " ";
        f++;
    }
}

cout << endl;
}

int main() {
    int ch, val;

    cout << "1) Insert\n";
    cout << "2) Delete\n";
    cout << "3) Display\n";
    cout << "4) Exit\n";

    do {
        cout << "Enter choice : ";
        cin >> ch;

```

```
switch (ch) {  
case 1:  
    cout << "Input for insertion: ";  
    cin >> val;  
    insertCQ(val);  
    break;  
  
case 2:  
    deleteCQ();  
    break;  
  
case 3:  
    displayCQ();  
    break;  
  
case 4:  
    cout << "Exit\n";  
    break;  
  
default:  
    cout << "Incorrect!\n";  
}  
  
} while (ch != 4);  
  
return 0;  
}
```

FILE NAME

circular_queue.cpp

COMMAND

g++ circular_queue.cpp -o circular_queue

./circular_queue